

ICLO x SNOWFLAKE

U.S. Self-Funded Employer Dental-Benefit PoC

Snowflake HLS GTM & Architecture Joint-Validation Proposal

August 2026

POSITIONING

This converts ICLO's existing Snowflake startup support into a structured 90-day validation.

EXECUTIVE DECISION BRIEF

A U.S. Self-Funded Employer Dental-Benefit PoC

Employees use dental benefits more easily; employers gain evidence for plan, administrator, access and workforce-benefit decisions.

1. Business fit Agree on the most natural Snowflake HLS sales play	2. Architecture Secure an HLS technical working session	3. Account evidence Validate one employer or administrator account team before introductions
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ICLO helps employees understand and use their dental benefits, navigate to plan-matching in-network options, and converts the journey into claims-verified, aggregate evidence for the employer. The current PoC scope is a U.S. self-funded employer dental benefit administered by a dental carrier or TPA.

Employer decision	Evidence needed
Plan + administrator fit	Eligibility quality, benefit rules, service levels and dental claims access
Network access	Workforce geography, provider overlap and real appointment availability
Utilization + cost allocation	Preventive completion; allowed, plan-paid and employee OOP
Evidence reliability	Denominator, source dates, claims lag, run-out and completeness
Workforce value	Participation, employee experience, turnover and multi-year scenarios

Short form: Employee dental-benefit navigation in the front; a governed evidence layer in the back.

CONFIRMED ICLO DIRECTION	Lead with employee dental-benefit navigation and employer aggregate evidence - not oral-image AI.
PRODUCT DESIGN PRINCIPLE	Keep model-derived signals separate from automated clinical routing; keep employee-level health signals out of employer views.
HYPOTHESIS / TO VALIDATE	The pattern can become repeatable across payer/TPA accounts that hold usable dental eligibility and claim-line data.
REQUEST TO SNOWFLAKE	Provide sales-play fit, named internal stakeholders, architecture decision support, account-team validation and written readiness criteria.

WHERE WE ACTUALLY ARE

What runs today, what is designed, and what does not exist yet

Read the rest of this document against this table. We do not describe design work as if it were running.

Stage	What	Evidence
Running today	An employer dashboard demo that computes synthetic numbers in the browser. No real data, no Snowflake connection yet.	Public demo
Designed	Evidence-layer data model, identity-resolution rules, policy and data-quality contracts, employer onboarding, inference API contract	46-page engineering design
To build in 90 days	The Snowflake evidence layer on real data, and a dashboard fed from it	Execution section
Does not exist	Employee web app, Core AI call path, benefits gateway. Separate track, not inside the 90 days.	Engineering design, sec. 8
Not yet decided	What the predictive model predicts, arm assignment, the full family and guardian flow	Engineering design, sec. 15 and 17
Blocking start	HIPAA role determination, BAA chain, employer-TPA data rights, legal basis for baseline load before consent	Start gates section
WHY THIS PAGE IS NEAR THE FRONT		The thicker a design document gets, the more it reads as a system already in production. One synthetic-data demo is running. Everything else is design and plan. Blur that line and we will each expect something different in 90 days.

MARKET CONTEXT + PROBLEM

Why the PoC Starts with a U.S. Self-Funded Employer Dental Benefit

Korea and the U.S. do not organize dental coverage in the same way; the target operating problem is specific to the U.S. employer-benefit lane.

	Korea: practical baseline	U.S. target context
Coverage structure	NHIS is the common public baseline; some dental services remain non-covered or out of pocket	Adult dental is often a separate benefit with its own plan, network and cost-sharing rules
Current PoC funding	Not the scope of this PoC	Self-funded employer bears claims risk; a dental carrier or TPA may administer
Employee difficulty	Determine covered vs. non-covered services and cost	Understand network, deductible, coinsurance, annual maximum and actual availability
Employer difficulty	Not the target operating model	Eligibility, plan, provider operations and dental claims sit across organizations
EMPLOYEE PROBLEM	Coverage does not automatically become usable care: plan rules, network economics, provider availability and final claims can differ from expectations.	
EMPLOYER PROBLEM	The employer needs evidence for plan and administrator fit, access gaps, utilization, cost allocation, claims completeness and workforce value.	
BOUNDARY	The comparison is directional. It does not claim that all Korean or all U.S. dental coverage follows one model.	

MARKET ECOSYSTEM

How the U.S. Dental Benefit Ecosystem Fits Together

Coverage segment, funding arrangement, administrator and beneficiary are separate questions; the current PoC covers one lane.

Market / funding lane	Who bears dental claim cost	Typical operating parties	Current PoC?
Employer fully insured	Dental carrier / insurer	Employer, broker, carrier, employees/dependents, dentists	No - market context
Employer self-funded / ASO	Employer	Employer, broker, dental carrier or TPA, employees/dependents, dentists	Yes - target lane
Individual / family	Individual and/or insurer under the product	Marketplace or direct issuer, member, dentist	No - market context
Medicaid / CHIP	Public program under federal/state structure	State, MCO where used, beneficiary, dentist	No - market context

Sponsor + broker Select funding, benefit design and administrator	Carrier / TPA / MCO Operate eligibility, benefits, network and/or claims	Member + dentist Receive care and submit dental claims	Claims + evidence Confirm procedures, allowed, plan-paid and OOP
CURRENT POC SCOPE			
U.S. self-funded employer + dental carrier/TPA + employees/dependents. Data rights, legal entity, cloud/region and claim-line availability must be validated account by account.			

PRODUCT

ICLO Connects Dental-Benefit Context to Employee-Initiated Action

ICLO is an employee benefits workflow with a claims-verified evidence layer - not another teledentistry consultation app.

Layer	What it contains	Operating discipline
Employee experience	Self-reported concern, benefit understanding, optional oral image, provider search, requested support	Employee action or human-assisted navigation remains explicit
Benefit and provider context	Eligibility, network, plan rules, deductible, coinsurance, annual maximum, available appointments	Only where contractually obtainable
Claims evidence	Member-month denominator, preventive completion, allowed, plan-paid, employee OOP, treatment mix, run-out	Employer output is aggregate only
Shadow signal	Image reference, capture time, model version, quality flag and signal band -- plus the person surrogate key, the capturing credential and the employer id, all of which are PHI-scoped	No raw disease probabilities; no automated provider, urgency or treatment-pathway decision

ICLO is not: a direct-to-consumer diagnostic AI, an underwriting or employee-risk scoring product, a provider-referral marketplace, or a first-year claims-savings promise. HomeDen PMS/EMR/CRM, insurance-product manufacturing, individual underwriting and automated claims adjudication are outside this PoC; they are conditional follow-on work, not part of what is being validated here.

PROVISIONAL REDLINE

Separate signal, navigation and referral-event tables; timestamp employee action and consent; maintain purpose-based access and model-version lineage, pending U.S. regulatory counsel review.

ARCHITECTURE

Snowflake as the Governed Evidence and Collaboration Plane

Structured eligibility, plan, event and dental-claims data converge in Snowflake; raw oral images follow a separate controlled path.

Structured evidence path

App events Consent and employee actions	HRIS eligibility Employee and dependent denominator	Plan + provider context Network and benefit rules	TPA/carrier claims Dental claim lines and run-out	Snowflake evidence layer Canonical model, controls, aggregate outcomes
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Raw-image path

Raw oral image Controlled U.S. object storage / PHI vault	Shadow inference Model endpoint and validation outside this proposal	Snowflake No image binaries. Person-linked references and signals do enter	Employer dashboard Aggregate views; no employee-level routing or health signal
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PLATFORM PREREQUISITE	Individual-level records exist in the governed processing layer - member-months, claim lines and app events cannot be computed without them. What the controls govern is who may read them: employer-facing output is aggregate and excludes employee-level PHI. If PHI is processed in Snowflake, Business Critical and an executed BAA are platform prerequisites. They do not make ICLO as a whole HIPAA-compliant.
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EVIDENCE DESIGN

The Dashboard Makes Data Quality and Privacy Visible

The current dashboard is synthetic visual proof, not a performance claim.

Visible control	Why the buyer needs it
Synthetic data - illustrative only	Prevents the demo from being mistaken for an ICLO outcome
Employees lens / members lens	Participation stands on plan-eligible employees at contracted employers (source: the eligibility feed supplied by the employer or its benefits administrator, typically X12 834) - the population, not the app users. Cost stands on covered member-months, which include enrolled dependents. The two denominators are never mixed.
Eligibility and claims dates	Shows which plan period and source cutoff each result uses
Claims lag and completeness	Makes run-out and missing-data uncertainty visible. The 60 days on screen is a demo constant; real lag is a distribution that varies by TPA, practice, procedure and season. A lag curve needs 12-18 months of history, so in pilot year one we show the claims-received-through date rather than an estimated completeness figure
Aggregate only; n >= 20 suppression	Provisional disclosure-control rule. The demo enforces it in browser JavaScript; the real system enforces it in the Snowflake query engine. Note that a Snowflake aggregation policy counts rows, not distinct people, unless an ENTITY KEY is set -- one person with 20 claim lines would otherwise satisfy a group of 20. It is not by itself a HIPAA de-identification determination
MESSAGE	This is not a performance claim. It demonstrates how denominator, freshness, privacy and claims completeness become visible to the buyer.

START GATES

No real data is loaded until these four are closed

These are contract and legal determinations, not engineering problems. They must be settled in writing before Day 30.

Gate	What has to be settled	If it stays open
HIPAA roles	Whether ICLO, the employer, the employer health plan, the TPA and Snowflake are each a covered entity, a business associate or a subcontractor. The BAA chain is built on top of that determination.	There is no lawful basis to load PHI.
Data rights	The data-rights clause in the employer-TPA agreement: by source, by field, by purpose, what we may receive and what we may use it for.	We receive data we cannot use. Most delay lands here.
Baseline load before consent	Baseline load precedes employee consent. At that moment nobody has consented, and in a pilot most people never will. Either load an aggregate-only baseline, or settle a separate legal basis in writing.	We would process the PHI of non-consenting employees without a basis. This is structural, not an edge case.
Common identity key	Whether a stable identifier exists on both the HR side and the TPA side to join a person across them.	Claims cannot be linked to app usage, so claims-confirmed completion does not hold.
THE ORDER MATTERS		Policy and access controls go in first, real data on top of them. Reversed, there is a window in which PHI exists without controls, and that window cannot be closed retroactively.

EXECUTION

A 90-Day Validation with Explicit Owners, Gates and Outputs

Snowflake and ICLO actions are separated by phase; ICLO owns the weekly cadence and one decision log.

Phase	Snowflake action	ICLO action	Gate / output
Step 0 + Days 0-30 Route + align	Route to U.S. HLS; confirm play, owners and account/BAA path	Provide current-state pack; run workshop and cadence	Sponsor, owners and route in writing
Days 31-60 Design	Size workload; co-draft security reference and account matrix	Provide source map, ingestion test and sizing inputs	Architecture and matrix v1 accepted
Days 61-90 Validate	Sponsor one account-team review; issue partner checklist	Run review; recommend go/no-go	Use case and evidence minimum agreed
Conditional Expand	Assess Native App, Clean Room or federated learning only if needed	Build only after multi-party demand is confirmed	Trigger: account review confirms multi-party need

OPERATING + CONSUMPTION MODEL

ICLO runs weekly written status and one decision log. ICLO supplies the sizing inputs - employers, member-months, ingestion/transform/outcome-refresh frequency, data volume, warehouse size and runtime - and the Snowflake SE calculates annual credits in the Day-30 sizing session. Day 0 is written sponsor acceptance; 'Meeting + N business days' and 'Day N' count from it.

We are not asking Snowflake to validate our dental AI or make ICLO compliant. We are asking Snowflake to help determine whether this can become a repeatable, governed payer/TPA data-collaboration pattern - and what ICLO must prove to make the partnership commercially real.

ECONOMICS

The Economics Start with Higher Year-One Claims - Not a Savings Promise

Preventive utilization and newly discovered treatment can increase dental plan-paid claims before treatment-mix change becomes measurable. The chart shows a cost path, not a return curve.

Period / segment	What may happen	How to interpret it
Year 1	Preventive visits rise; untreated conditions are discovered	Allowed, plan-paid and employee OOP may move differently
Years 2-3	Treatment mix and claims run-out become more observable	Direction remains a customer-specific hypothesis
Low turnover	More time for the employer to observe multi-year effects	A multi-year value case may be more relevant
High turnover	Untreated need may re-enter with new hires	Employee experience, recruitment and retention may be the stronger case

ICLO will not contract on a year-one savings guarantee.

Planning inputs include population age, geography, network overlap, provider availability, turnover, plan design, historical claims, paid/OOP allocation, untreated-need assumptions and claims run-out. Outputs include Year 1-3 scenarios, turnover sensitivity, network-access sensitivity and explicit uncertainty.

HYPOTHESIS	Simulation values are planning heuristics to be calibrated with customer data - not actual ICLO outcomes.
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RESPONSIBILITIES

What Snowflake Can Support - and What Remains Outside Its Scope

The platform can enable governed evidence and collaboration; it does not decide ICLO's intended use, data rights or legal sufficiency.

Snowflake can support	ICLO / customer responsibility	Outside the Snowflake ask
Governed storage and compute; canonical model; reconciliation	Source mapping, product logic and operational controls	Employer/TPA data-rights determination
Role-based access, masking, row policies and access history	Consent UX, purpose metadata and incident operations	Legal sufficiency of employee consent
Aggregate outcome views and claims-confirmed calculations	Program attribution logic and customer reporting	First-year savings validation
Same-region share where available; controlled staging otherwise	Connector and source-system implementation	Cross-region legal conclusions
Workload sizing; pre-credit run-rate economics	Commercial assumptions and usage monitoring	FDA status, licensure or referral-law analysis; ICLO-wide HIPAA compliance conclusion

Native App, multi-party Clean Room and federated learning are conditional options. They are not prerequisites for the first employer pilot.

PARTNERSHIP REQUEST

Convert Existing Snowflake Support into a Routed Validation

Step 0 is routing: the current Korea startup sponsor names the U.S. HLS GTM and architecture owners before technical work begins.

Priority	Requested output
Step 0 / immediate	Official support-program name and current sponsor; routed U.S. HLS GTM and architecture owners; U.S. account/region/Business Critical/BAA path; written credits and support scope
Next	Self-funded employer dental and administrator/TPA account map; one account-team use-case validation; Direct Share versus SFTP/API decision pattern; partner checklist; co-sell milestones
Later / conditional	Native App feasibility; multi-party Clean Room; federated-learning feasibility; joint case study; optional Snowflake employee-benefits design partnership, subject to fit

- Which Snowflake HLS sales play fits most naturally?
- Is the stronger Snowflake story the employee application or the governed payer/TPA evidence layer behind it?
- What must ICLO prove before an account team would actively introduce or co-sell?
- Should the path be Snowflake for Startups, Partner Network, Powered by Snowflake or an ISV/Native App route?

ABOUT ICLO

Confirmed Facts and Fields to Complete Before Release

ICLO has defined the product/data boundary and target PoC lane; company and Snowflake account fields must be completed before external release.

Field	Current statement
Company	ICLO Co., Ltd. · Jeju, Republic of Korea · Commercial stage in the Korean market (B2C and B2B) · 12 core team members
Product	HomeDen SaaS (PMS / EMR / CRM) plus employer dental-benefit navigation and claims-evidence design
Program selections	Selected: Fintech Cube Cohort 9 and Shinhan Future's Lab Cohort 12
Snowflake relationship	Existing Snowflake startup support
Current Snowflake state	[Account, cloud, region, credits, usage and technical-support owner: confirm]
Next proof	One U.S. self-funded employer dental-data path and one account-team use-case review
RELEASE GATE	The remaining bracketed fields are the ask in this meeting, not gaps in ICLO's own record. Korean commercial stage and program selection do not prove insurer integration, underwriting, adjudication or U.S. outcomes.

REFERENCES AND REVIEW

Source Notes and Self-Check

External facts are separated from ICLO direction, product principles, hypotheses and requests.

- U.S. Department of Labor, EBSA - Understanding Your Fiduciary Responsibilities Under a Group Health Plan: an ERISA-covered group health plan may cover dental, and may fund benefits through a trust, by purchasing insurance, or by self-funding from the employer's general assets.
<https://www.dol.gov/sites/dolgov/files/EBSA/about-ebsa/our-activities/resource-center/publications/group-health-plan-fiduciary-responsibilities.pdf> (section page:
<https://www.dol.gov/agencies/ebsa/employers-and-advisers/plan-administration-and-compliance/health-plans>)
- U.S. Department of Labor, EBSA - Annual Report to Congress on Self-Insured Group Health Plans (2025 edition, the current one): self-insured sponsor pays claims directly; fully insured sponsor pays premiums and the insurer assumes risk.
<https://www.dol.gov/sites/dolgov/files/EBSA/researchers/statistics/retirement-bulletins/annual-report-on-self-insured-group-health-plans-2025.pdf> (all editions: <https://www.dol.gov/agencies/ebsa/about-ebsa/our-activities/resource-center/reports>)
- HealthCare.gov - adult dental is not an essential health benefit and may be embedded or separate:
<https://www.healthcare.gov/coverage/dental-coverage/>
- Medicaid.gov, preventive services - pediatric/CHIP dental requirements and state-option adult dental:
<https://www.medicaid.gov/medicaid/benefits/prevention>
- American Dental Association - dental benefit plan designs:
<https://www.ada.org/resources/practice/dental-insurance/benefit-plan-designs>
- Korea NHIS - national health insurance benefit framework:
<https://www.nhis.or.kr/english/wbheaa02600m01.do>
- Snowflake Editions / Business Critical: <https://docs.snowflake.com/en/user-guide/intro-editions>
- Snowflake row access policies: <https://docs.snowflake.com/en/user-guide/security-row-intro>
- Snowflake data protection policies: <https://docs.snowflake.com/en/user-guide/data-protection-policies-snowsight>
- Snowflake Access History: <https://docs.snowflake.com/en/user-guide/access-history>
- Cross-region sharing and replication: <https://docs.snowflake.com/en/user-guide/secure-data-sharing-across-regions-platforms>
- Snowflake Native App Framework: <https://docs.snowflake.com/en/developer-guide/native-apps/native-apps-about>
- Snowflake Data Clean Rooms: <https://docs.snowflake.com/en/user-guide/cleanrooms/about>

Self-check: the U.S. self-funded employer dental PoC is explicit from the opening; ICLO is not presented as teledentistry; coverage, funding and administration are separated; Snowflake's role goes beyond hosting; employer/TPA/carrier/employee roles are distinct; no first-year savings is promised; raw-image and structured-data boundaries are clear; Business Critical is not framed as automatic HIPAA compliance; federated learning is conditional; named owners and outputs remain at close.